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RESEARCH ARTICLE

Metamorphic Testing for Semantic Invariance in Large Language Models

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ABSTRACT Large Language Models (LLMs) have demonstrated remarkable capabilities in mathematical and scientific reasoning tasks, yet standard benchmark evaluations fail to capture the stability of these reasoning processes under semantically equivalent input variations. We introduce a metamorphic testing framework for assessing semantic invariance in LLM reasoning, systematically evaluating model responses across eight metamorphic relations: identity, paraphrase, fact reordering, expansion, contraction, academic context, business context, and contrastive formulation. We conduct a comparative analysis of two recent foundation models—Hermes-4-70B and DeepSeek-R1-0528—across 79 reasoning problems spanning eight categories (Physics, Mathematics, Chemistry, Economics, Statistics, Biology, Calculus, and Optimization) at three difficulty levels. Both models exhibit similar aggregate invariance scores (∼0.90), yet manifest distinct vulnerability profiles–DeepSeek shows notable sensitivity to paraphrase and fact-reordering transformations, particularly in formal domains (Calculus: -0.104, Statistics: -0.048). Contrastive transformations prove particularly challenging for Hermes, inducing mean score deltas of -0.147 (Hermes) and -0.005 (DeepSeek). Semantic similarity analysis further reveals that Hermes maintains more coherent reasoning traces (mean similarity: 0.87) compared to DeepSeek (0.78), with DeepSeek exhibiting near-complete reasoning breakdown in specific category-transformation combinations. These findings demonstrate that metamorphic testing uncovers robustness characteristics invisible to conventional accuracy metrics, providing essential insights for deploying LLMs in high-stakes reasoning applications. We release our evaluation framework and complete experimental data to facilitate reproducible robustness assessment of foundation models.

INDEX TERMS Metamorphic testing, semantic invariance, large language models, reasoning robustness, LLM evaluation.

I. INTRODUCTION

Large Language Models (LLMs) have achieved unprecedented performance across a wide spectrum of reasoning tasks, from mathematical problem-solving to scientific question answering [1], [2], [3]. These advances have catalyzed their deployment in increasingly consequential domains [4], including educational assessment [5], scientific discovery [6], and decision support systems [7]. However, the standard evaluation paradigm—measuring accuracy on curated benchmark datasets—provides an incomplete picture of model capabilities, failing to capture a fundamental desideratum for reliable reasoning systems: *semantic invariance*.

A robust reasoning agent should produce consistent outputs when presented with semantically equivalent inputs. A physics problem asking about projectile motion should yield the same solution whether phrased in formal academic language or practical business terminology, whether the given

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facts appear in their original order or are permuted, and whether the problem statement is expanded with clarifying context or contracted to its essential elements. Yet, recent investigations have revealed that LLMs exhibit surprising brittleness to superficial input perturbations that preserve semantic content [8], [9], [10]. This fragility raises critical concerns about the reliability of LLM reasoning in real-world deployments, where input formulations are inherently variable and uncontrolled.

Metamorphic testing [11], originally developed for validating software systems lacking test oracles, offers a principled framework for assessing this dimension of model robustness. Rather than requiring ground-truth labels for transformed inputs, metamorphic testing specifies *metamorphic relations* (MRs)—expected relationships between outputs given known relationships between inputs. For semantically equivalent transformations, the metamorphic relation is straightforward: the model should produce equivalent reasoning and conclusions. Violations of this invariance property indicate reasoning instability that standard accuracy metrics cannot detect.

In this work, we introduce a comprehensive metamorphic testing framework for evaluating semantic invariance in LLM reasoning. Our framework operationalizes eight metamorphic relations spanning three categories of semantic-preserving transformations:

- **Structural transformations**: *identity* (baseline), *paraphrase* (lexical and syntactic variation), and *reorder_facts* (permutation of given information);
- **Verbosity transformations**: *expand* (addition of clarifying but non-informative content) and *contract* (removal of redundant linguistic material);
- **Contextual transformations**: *context_academic* (formal scholarly framing), *context_business* (practical professional framing), and *contrastive* (explicit contrast with alternative scenarios).

We apply this framework to conduct a systematic comparative analysis of two recent foundation models—Hermes-4-70B [12] and DeepSeek-R1-0528 [13]—representing distinct architectural lineages and training methodologies. Our evaluation encompasses 79 reasoning problems distributed across eight subject categories (Physics, Mathematics, Chemistry, Economics, Statistics, Biology, Calculus, and Optimization) at three difficulty levels (Easy, Medium, Hard), yielding a total of 632 metamorphic test instances per model.

Our analysis reveals several findings that challenge conventional understanding of LLM reasoning capabilities:

1) **Accuracy-stability dissociation**: We observe a striking disconnect between reasoning accuracy and process fidelity. Hermes-4-70B achieves substantially higher overall reasoning scores (0.577 vs. 0.425), yet DeepSeek-R1-0528 demonstrates superior step-level accuracy (0.640 vs. 0.531). This dissociation suggests fundamentally different reasoning strategies that aggregate metrics obscure.
2) **Transformation-specific vulnerabilities**: While both models achieve similar aggregate invariance scores ($\sim$ 0.90), they exhibit markedly different vulnerability profiles. DeepSeek-R1-0528 shows catastrophic sensitivity to paraphrase and fact-reordering transformations, particularly in formal domains such as Calculus (score delta: -0.104) and Statistics (-0.048). Hermes-4-70B demonstrates more uniform, moderate degradation across transformations.
3) **Universal contrastive fragility**: Contrastive transformations—presenting problems alongside explicit contrasts with alternative scenarios—prove universally destabilizing, inducing the largest mean score degradation for Hermes-4-70B (-0.147) and substantial variance for both models. This finding suggests that LLMs struggle to maintain reasoning focus when presented with potentially distracting comparative information.
4) **Semantic coherence divergence**: Analysis of reasoning trace similarity reveals that Hermes-4-70B maintains more coherent step-by-step reasoning under transformation (mean semantic similarity: 0.87) compared to DeepSeek-R1-0528 (0.78), with the latter exhibiting near-complete reasoning breakdown in specific category-transformation combinations.

These findings carry important implications for both the evaluation and deployment of LLM reasoning systems. From an evaluation perspective, our results demonstrate that metamorphic testing uncovers robustness characteristics invisible to conventional accuracy-based assessment, providing a complementary lens for model comparison. From a deployment perspective, the identified vulnerability patterns offer actionable guidance for input preprocessing and prompt engineering strategies to mitigate reasoning instability.

A. CONTRIBUTIONS

The principal contributions of this work are:

1) A comprehensive metamorphic testing framework for semantic invariance evaluation in LLM reasoning, operationalizing eight metamorphic relations across structural, verbosity, and contextual transformation categories;
2) An extensive empirical analysis comparing two foundation models across 1,264 metamorphic test instances, revealing systematic patterns of reasoning instability that standard benchmarks fail to capture;
3) Detailed characterization of transformation-specific and category-specific vulnerability profiles, providing actionable insights for robust LLM deployment;
4) Public release of our evaluation framework, transformed problem sets, and complete experimental results to facilitate reproducible robustness assessment of foundation models.[1]

[1]Code and data available at: https://github.com/drdezarza/metamorphic-llm-evaluation

TABLE 1. Comparison of our approach with related evaluation methodologies. Columns indicate: evaluation of semantic invariance (SI), multiple metamorphic relations (MR), reasoning trace analysis (RT), multi-category evaluation (MC), and difficulty stratification (DS).

Approach	SI	MR	RT	MC	DS
MMLU [22]	✗	✗	✗	✓	✗
PromptBench [9]	✗	✓	✗	✓	✗
CheckList [43]	✓	✓	✗	✗	✗
ParaRel [31]	✓	✗	✗	✗	✗
GSM-IC [10]	✓	✗	✗	✗	✗
Ours	✓	✓	✓	✓	✓

B. PAPER ORGANIZATION

The remainder of this paper is structured as follows. Section II reviews related work on LLM robustness evaluation and metamorphic testing. Section III details our metamorphic testing framework, including the definition of metamorphic relations and evaluation metrics. Section IV describes the experimental setup, including model selection, problem corpus construction, and transformation procedures. Section V presents our empirical findings, organized by overall performance, metamorphic relation analysis, category-specific patterns, and difficulty-stratified results. Section VI concludes.

II. RELATED WORK

Our work intersects several active research areas: LLM evaluation methodologies, robustness assessment, metamorphic testing, and semantic consistency analysis. We review each in turn, positioning our contributions relative to existing approaches.

A. LLM EVALUATION AND REASONING BENCHMARKS

The rapid advancement of large language models has spurred the development of increasingly sophisticated evaluation benchmarks. Early benchmarks focused on narrow linguistic competencies [14], [15], while subsequent efforts expanded to assess broader reasoning capabilities. Mathematical reasoning benchmarks such as GSM8K [16], MATH [17], and MathBench [18] evaluate multi-step quantitative problem-solving. Scientific reasoning is assessed through benchmarks including SciQ [19], ARC [20], and ScienceQA [21]. Comprehensive evaluation suites such as MMLU [22], BIG-Bench [23], and HELM [24] attempt to provide holistic assessments across diverse task categories.

Despite their breadth, these benchmarks share a common limitation: they evaluate models on fixed, canonical problem formulations, implicitly assuming that performance generalizes across semantically equivalent phrasings. This assumption has been repeatedly challenged by demonstrations of LLM sensitivity to superficial input variations [25], [26], [27]. Our work directly addresses this gap by systematically evaluating reasoning stability under controlled semantic-preserving transformations.

Recent efforts have begun incorporating robustness considerations into benchmark design. PromptBench [9] evaluates LLM resilience to adversarial prompt perturbations, while RobustQA [28] assesses question-answering stability. However, these approaches typically focus on adversarial perturbations designed to degrade performance, rather than semantic-preserving transformations that should ideally yield invariant outputs. Our metamorphic testing framework is complementary, specifically targeting the semantic invariance property that robust reasoning systems should exhibit.

B. ROBUSTNESS AND CONSISTENCY IN LARGE LANGUAGE MODELS

The study of LLM robustness encompasses multiple dimensions, including adversarial robustness [8], [29], out-of-distribution generalization [30], and consistency under paraphrase [31], [32]. We focus on the latter, which most directly relates to semantic invariance.

Reference [31] introduced the notion of *consistency* as a desideratum for language models, demonstrating that BERT-family models frequently produce contradictory predictions for paraphrased factual queries. Reference [32] extended this analysis to evaluate systematic generalization, finding that models often fail to transfer knowledge across superficially different formulations of equivalent problems. Reference [10] showed that LLMs exhibit significant performance degradation when irrelevant context is added to mathematical word problems, suggesting sensitivity to verbosity transformations.

Self-consistency decoding [33] represents an orthogonal approach, leveraging multiple sampling paths to improve reasoning reliability through majority voting. While effective at improving aggregate accuracy, this technique does not address the underlying instability of individual reasoning traces, which our analysis directly characterizes.

Recent work has also examined the consistency of chain-of-thought reasoning [34]. Reference [35] demonstrated that LLMs often produce unfaithful reasoning traces where stated intermediate steps do not causally influence final answers. Reference [36] showed that chain-of-thought explanations can be systematically biased by irrelevant features. Our semantic similarity analysis of reasoning traces extends this line of inquiry, quantifying how transformation-induced perturbations propagate through multi-step reasoning processes.

C. METAMORPHIC TESTING

Metamorphic testing, introduced by [37], addresses the test oracle problem in software verification—the challenge of determining expected outputs when ground truth is unavailable or expensive to obtain. Rather than specifying expected outputs directly, metamorphic testing defines *metamorphic relations* (MRs) that constrain relationships between inputs and outputs across multiple executions. If a system violates

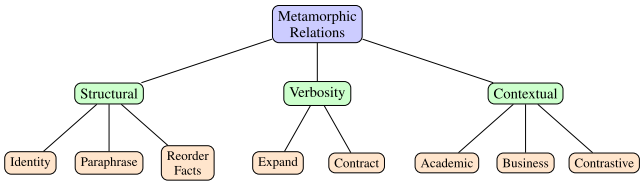


FIGURE 1. Taxonomy of metamorphic relations for semantic invariance evaluation. Eight MRs are organized into three categories: structural transformations (affecting linguistic form), verbosity transformations (affecting length/detail), and contextual transformations (affecting framing).

an MR, a fault is detected without requiring explicit ground truth for transformed inputs.

The approach has been successfully applied across diverse domains, including numerical computation [11], computer graphics [38], machine learning systems [39], [40], and autonomous vehicles [41], [42]. For machine learning specifically, [39] proposed metamorphic relations based on data permutation invariance, while [40] developed MRs for testing image classification systems.

Application of metamorphic testing to natural language processing systems remains relatively nascent. Reference [43] introduced CheckList, which includes invariance tests as one component of behavioral testing for NLP models. Reference [44] proposed metamorphic testing for detecting fairness violations in NLP models, defining MRs based on linguistic transformations. Reference [45] proposed a property-based method for validating machine reading comprehension software without requiring annotated labels, testing systems by checking behavioral consistency across related queries.

Our work advances this line of research by developing a comprehensive metamorphic testing framework specifically designed for evaluating reasoning processes in LLMs. Unlike prior approaches that focus primarily on final answer correctness, we analyze both solution quality and reasoning trace coherence, providing a more nuanced characterization of semantic invariance.

D. SEMANTIC SIMILARITY AND TEXTUAL ENTAILMENT

Quantifying semantic equivalence between text passages is fundamental to our evaluation methodology. Approaches to semantic similarity have evolved from lexical overlap metrics [46], [47] to embedding-based methods [48], [49] and learned entailment models [50], [51].

Sentence embedding models, particularly those based on contrastive learning objectives [49], [52], have demonstrated strong performance on semantic textual similarity benchmarks. These models project sentences into a shared embedding space where cosine similarity correlates with human judgments of semantic relatedness. We employ such embeddings to quantify reasoning trace similarity across original and transformed problem instances.

Natural language inference (NLI) models trained on datasets such as SNLI [50] and MultiNLI [51] provide an alternative approach, predicting entailment, contradiction, or neutrality relationships between premise-hypothesis pairs. While NLI models offer fine-grained relationship classification, embedding-based similarity provides continuous-valued scores more amenable to aggregate statistical analysis. Our framework employs embedding-based similarity as the primary metric, though NLI-based analysis represents a promising direction for future work.

E. POSITIONING OF OUR WORK

Our work synthesizes insights from the above research threads into a unified framework for evaluating semantic invariance in LLM reasoning. We make several distinct contributions relative to prior work:

- **Comprehensive MR taxonomy**: While prior metamorphic testing work for NLP has employed limited transformation sets, we systematically operationalize eight MRs spanning structural, verbosity, and contextual categories, enabling fine-grained analysis of transformation-specific vulnerabilities.
- **Reasoning process analysis**: Beyond final answer evaluation, we analyze reasoning trace coherence through semantic similarity metrics, characterizing how perturbations propagate through multi-step reasoning.
- **Multi-dimensional stratification**: Our evaluation is stratified across problem categories and difficulty levels, revealing interaction effects between transformation type and problem characteristics that aggregate analyses obscure.
- **Comparative model analysis**: By evaluating multiple models under identical conditions, we identify model-specific vulnerability profiles, providing insights into how different training methodologies and architectures influence reasoning robustness.

Table 1 summarizes the positioning of our work relative to representative prior approaches.

III. METHODOLOGY

This section presents our metamorphic testing framework for evaluating semantic invariance in LLM reasoning. We begin with a formal problem statement, then detail the metamorphic relations employed, and conclude with the evaluation metrics used to quantify invariance and reasoning quality.

A. PROBLEM FORMULATION

Let $\mathcal{M}$ denote a large language model that maps a problem statement $p \in \mathcal{P}$ to a solution $s = \mathcal{M}(p)$, where s comprises both a final answer a and a reasoning trace $r = (r_1, r_2, \ldots, r_k)$ consisting of k intermediate steps. We consider the setting where each problem p has an associated reference solution $s^* = (a^*, r^*)$ against which model outputs are evaluated.

A *semantic-preserving transformation* $\tau : \mathcal{P} \to \mathcal{P}$ maps a problem p to a transformed variant $p' = \tau(p)$ such that the semantic content relevant to solving the problem is preserved.

Formally, we require that any correct solution to p remains correct for p', and vice versa. The set of such transformations defines an equivalence relation over problem statements.

Definition 1 (Semantic Invariance): A model $\mathcal{M}$ exhibits *perfect semantic invariance* with respect to transformation τ if, for all problems $p \in \mathcal{P}$:

$$\mathcal{M}(p) \equiv \mathcal{M}(\tau(p)) \tag{1}$$

where $\equiv$ denotes semantic equivalence of solutions.

In practice, perfect invariance is neither expected nor required. Instead, we seek to quantify the degree to which a model's outputs remain stable under semantic-preserving transformations, identifying systematic patterns of instability that may indicate reasoning fragility.

B. METAMORPHIC RELATIONS

We operationalize eight metamorphic relations (MRs) organized into three categories based on the nature of the transformation. Each MR defines a specific semantic-preserving transformation along with the expected invariance property. Figure 1 illustrates the taxonomy, and Table 2 provides formal definitions with examples.

1) STRUCTURAL TRANSFORMATIONS

Structural transformations modify the linguistic form or organization of problem statements while preserving informational content.

a: IDENTITY (τ_{ID})

The identity transformation returns the original problem unchanged: $\tau_{\text{id}}(p) = p$. This serves as a baseline for quantifying intrinsic model variance due to sampling stochasticity, establishing a floor against which other transformations are compared.

b: PARAPHRASE (τ_{PARA})

The paraphrase transformation rephrases the problem statement using alternative lexical choices and syntactic structures while preserving meaning:

$$\tau_{\text{para}}(p) = \text{Paraphrase}(p) \quad \text{s.t.} \quad \text{sem}(p) = \text{sem}(\tau_{\text{para}}(p)) \tag{2}$$

where $\text{sem}(\cdot)$ denotes the semantic content function. Paraphrasing is implemented via prompted LLM generation with explicit instructions to preserve all quantitative values, variable names, and logical relationships.

c: REORDER FACTS (τ_{REORD})

Many reasoning problems present multiple facts or constraints that can be stated in various orders. The fact reordering transformation permutes the presentation order of independent facts:

$$\tau_{\text{reord}}(p) = \text{Permute}(\text{Facts}(p)) \tag{3}$$

where $\text{Facts}(p)$ extracts the set of independent factual statements from p. A robust reasoner should be invariant to presentation order when facts are logically independent.

2) VERBOSITY TRANSFORMATIONS

Verbosity transformations modify the length and detail level of problem statements without altering core informational content.

a: EXPAND (τ_{EXP})

The expansion transformation augments the problem statement with additional clarifying context, definitions, or elaborations that do not provide new information necessary for solving the problem:

$$\tau_{\text{exp}}(p) = p \oplus \text{Elaboration}(p) \tag{4}$$

where $\oplus$ denotes concatenation with appropriate integration. Expansions may include restatements of given information, definitional clarifications, or contextual framing. This transformation tests robustness to verbose inputs where essential information is embedded within supplementary context.

b: CONTRACT (τ_{CON})

The contraction transformation removes redundant or non-essential linguistic material while preserving all information necessary to solve the problem:

$$\tau_{\text{con}}(p) = \text{Minimize}(p) \quad \text{s.t.} \quad \text{Info}(p) \subseteq \text{Info}(\tau_{\text{con}}(p)) \tag{5}$$

This transformation tests whether models can extract essential information from concise formulations, or whether they rely on verbose cues present in typical training data.

3) CONTEXTUAL TRANSFORMATIONS

Contextual transformations embed the problem within different framing contexts that should not affect the reasoning process or solution.

a: ACADEMIC CONTEXT (τ_{ACAD})

The academic context transformation frames the problem within a scholarly or educational setting:

$$\tau_{\text{acad}}(p) = \text{Frame}_{\text{academic}}(p) \tag{6}$$

This may include framing the problem as an exam question, textbook exercise, or research scenario. The transformation tests whether models exhibit context-dependent reasoning biases.

b: BUSINESS CONTEXT (τ_{BUS})

The business context transformation frames the problem within a professional or commercial setting:

$$\tau_{\text{bus}}(p) = \text{Frame}_{\text{business}}(p) \tag{7}$$

For instance, a physics problem about velocity might be reframed as a logistics optimization scenario. This tests whether models maintain reasoning quality when problems are presented in applied rather than abstract contexts.

c: CONTRASTIVE (τ_{CONTR})

The contrastive transformation augments the problem with explicit contrasts to alternative scenarios or common misconceptions:

$$\tau_{\text{contr}}(p) = p \oplus \text{Contrast}(p) \tag{8}$$

For example, a problem might be augmented with statements like "Note that this differs from the case where..." or "Unlike problems involving X, here we have Y." This tests whether models can maintain focus on the actual problem when presented with potentially distracting comparative information.

C. EVALUATION METRICS

We employ a multi-faceted evaluation framework that assesses both solution quality and reasoning process coherence. Our metrics operate at three levels: solution-level, step-level, and trace-level.

1) SOLUTION-LEVEL METRICS

a: SEMANTIC SIMILARITY SCORE

Given a model solution s and reference solution s^*, we compute the semantic similarity using embedding-based comparison:

$$\text{Score}(s, s^*) = \cos(\mathbf{e}_s, \mathbf{e}_{s^*}) = \frac{\mathbf{e}_s \cdot \mathbf{e}_{s^*}}{\|\mathbf{e}_s\| \|\mathbf{e}_{s^*}\|} \tag{9}$$

where $\mathbf{e}_s$ and $\mathbf{e}_{s^*}$ are sentence embeddings obtained from a pretrained encoder model. We employ the `all-MiniLM-L6-v2` model from the Sentence-Transformers library [48], which provides a good balance between computational efficiency and semantic fidelity.

b: SCORE DELTA

The primary invariance metric is the score delta, measuring the change in solution quality under transformation:

$$\Delta_\tau(p) = \text{Score}(\mathcal{M}(\tau(p)), s^*) - \text{Score}(\mathcal{M}(p), s^*) \tag{10}$$

Negative values indicate performance degradation under transformation, positive values indicate improvement, and values near zero indicate invariance. For a perfectly invariant model, $\mathbb{E}[\Delta_\tau] = 0$ for all semantic-preserving transformations τ.

2) STEP-LEVEL METRICS

a: STEP SCORE

For problems with multi-step reference solutions $r^* = (r_1^*, \ldots, r_k^*)$, we evaluate each step of the model's reasoning trace:

$$\text{StepScore}_o(r, r^*) = \max_z \cos(\mathbf{e}_{r_o}, \mathbf{e}_{r_z^*}) \tag{11}$$

This formulation allows for reordering of reasoning steps, acknowledging that valid solution paths may traverse steps in different orders.

b: AVERAGE STEP SCORE

The aggregate step-level accuracy is computed as:

$$\text{AvgStepScore}(r, r^*) = \frac{1}{k} \sum_{o=1}^{k} \text{StepScore}_o(r, r^*) \tag{12}$$

c: STEP DELTA

Analogous to the solution-level score delta, we compute step-level deltas:

$$\Delta_\tau^{\text{step}}(p) = \text{AvgStepScore}(r_{\tau(p)}, r^*) - \text{AvgStepScore}(r_p, r^*) \tag{13}$$

3) TRACE-LEVEL METRICS

a: SEMANTIC SIMILARITY OF REASONING TRACES

To quantify the coherence of reasoning processes across transformations, we measure the similarity between reasoning traces produced for original and transformed problems:

$$\text{TraceSim}_\tau(p) = \cos(\mathbf{e}_{r_p}, \mathbf{e}_{r_{\tau(p)}}) \tag{14}$$

where $\mathbf{e}_r$ denotes the embedding of the concatenated reasoning trace. High trace similarity indicates that the model follows similar reasoning paths regardless of surface-level problem formulation.

4) AGGREGATE ROBUSTNESS METRICS

We define several aggregate metrics to characterize overall model robustness:

a: MEAN ABSOLUTE DELTA (MAD)

The average magnitude of score change across all transformations:

$$\text{MAD} = \frac{1}{|\mathcal{T}||\mathcal{P}|} \sum_{\tau \in \mathcal{T}} \sum_{p \in \mathcal{P}} |\Delta_\tau(p)| \tag{15}$$

where $\mathcal{T}$ is the set of non-identity transformations. Lower MAD indicates greater invariance.

b: INVARIANCE SCORE

A normalized robustness measure:

$$\text{InvScore} = 1 - \text{MAD} \tag{16}$$

Values closer to 1 indicate stronger semantic invariance.

c: DEGRADATION RATE

The proportion of transformation instances resulting in significant performance degradation:

$$\text{DegRate}_\theta = \frac{1}{|\mathcal{T}||\mathcal{P}|} \sum_{\tau \in \mathcal{T}} \sum_{p \in \mathcal{P}} \mathbb{I}[\Delta_\tau(p) < -\theta] \tag{17}$$

where θ is a significance threshold (we use $\theta = 0.1$) and $\mathbb{I}[\cdot]$ is the indicator function.

TABLE 2. Metamorphic relation definitions with illustrative examples. Each transformation preserves the semantic content necessary for problem solution.

MR	Original	Transformed
Identity	A train travels 120 km in 2 hours. What is its average speed?	A train travels 120 km in 2 hours. What is its average speed?
Paraphrase	A train travels 120 km in 2 hours. What is its average speed?	A locomotive covers a distance of 120 kilometers over a 2-hour period. Calculate the mean velocity.
Reorder Facts	A ball is thrown upward at 20 m/s. Gravity is 9.8 m/s^2. Find max height.	Gravity is 9.8 m/s^2. A ball is thrown upward at 20 m/s. Find max height.
Expand	Solve: $2x^2 - 5x - 12 = 0$	Solve the following quadratic equation, where we seek values of the variable x that satisfy the equality: $2x^2 - 5x - 12 = 0$
Contract	Calculate the average speed of a train that travels a total distance of 120 km in 2 hours.	Train: 120 km, 2 hours. Average speed?
Academic	Find the pH of 0.01 M HCl.	[Exam Question] In a chemistry laboratory exercise, determine the pH of a 0.01 M HCl solution.
Business	Find the pH of 0.01 M HCl.	A pharmaceutical company needs to verify the acidity of a 0.01 M HCl solution for quality control. Calculate the pH.
Contrastive	Find max height for a ball thrown at 20 m/s.	Find max height for a ball thrown at 20 m/s. Note: unlike horizontal projectile motion, vertical motion involves continuous deceleration.

d: CONSISTENCY

Following [31], we measure output consistency as the standard deviation of scores across transformation variants:

$$\text{Consistency}(p) = 1 - \text{Std}_{\tau \in \mathcal{T}}[\text{Score}(\mathcal{M}(\tau(p)), s^*)] \quad (18)$$

D. STATISTICAL ANALYSIS

To assess the significance of observed differences, we employ non-parametric statistical tests appropriate for our evaluation setting:

1) BETWEEN-MODEL COMPARISONS

We use the Mann-Whitney U test [53] to compare score delta distributions between models, as these distributions are not assumed to be normally distributed.

2) WITHIN-MODEL MR EFFECTS

The Kruskal-Wallis H test [54] is used to assess whether score deltas differ significantly across metamorphic relations within a single model.

3) EFFECT SIZE

We report Cohen's d [55] to quantify the practical significance of observed differences:

$$d = \frac{\mu_1 - \mu_2}{s_{\text{pooled}}} \quad (19)$$

where s_{pooled} is the pooled standard deviation. Following conventional thresholds, $|d| < 0.5$ indicates a small effect, $0.5 \leq |d| < 0.8$ a medium effect, and $|d| \geq 0.8$ a large effect.

4) CORRELATION ANALYSIS

Pearson and Spearman correlations are computed to examine relationships between metrics, such as the association between score delta and semantic trace similarity.

IV. EXPERIMENTAL SETUP

This section describes the experimental configuration used to evaluate semantic invariance in LLM reasoning. We detail model selection criteria, problem corpus construction, transformation implementation, inference configuration, and the evaluation pipeline.

A. MODEL SELECTION

We evaluate two recent large language models representing distinct development lineages and training methodologies:

1) HERMES-4-70B

Hermes-4-70B [12] is a 70-billion parameter model developed by NousResearch, fine-tuned from the LLaMA architecture [3]. The model emphasizes instruction-following capabilities and has demonstrated strong performance on reasoning benchmarks. Hermes models are trained with a focus on helpful, harmless, and honest responses, incorporating reinforcement learning from human feedback (RLHF) and direct preference optimization (DPO) techniques.

TABLE 3. Specifications of evaluated models.

Specification	Hermes-4-70B	DeepSeek-R1-0528
Parameters	70B	70B (active)
Architecture	Dense Transformer	Mixture-of-Experts
Base Model	LLaMA	DeepSeek-V2
Context Length	8,192 tokens	32,768 tokens
Training Focus	Instruction-following	Reasoning optimization
Release Date	August 26, 2025	May 28, 2025

2) DEEPSEEK-R1-0528

DeepSeek-R1-0528 [13] is developed by DeepSeek AI, representing their reasoning-optimized model series. The R1 series is specifically designed for enhanced reasoning capabilities, incorporating extended chain-of-thought training and reasoning-specific data curation. The model employs a mixture-of-experts (MoE) architecture, enabling efficient scaling of model capacity.

These models were selected based on several criteria: (1) comparable scale enabling fair comparison, (2) distinct training methodologies providing diversity in reasoning approaches, (3) strong baseline performance on standard reasoning benchmarks, and (4) public availability enabling reproducibility. Table 3 summarizes key model specifications.

B. PROBLEM CORPUS

We use a diverse corpus of reasoning problems [56] spanning multiple domains and difficulty levels to enable comprehensive evaluation of semantic invariance.

1) CATEGORY COVERAGE

The corpus encompasses eight subject categories, selected to represent a breadth of reasoning types:

- **Physics** ($n = 11$): Kinematics, dynamics, thermodynamics, and wave mechanics problems requiring application of physical laws and quantitative reasoning.
- **Mathematics** ($n = 11$): Algebraic manipulation, equation solving, and number theory problems testing formal mathematical reasoning.
- **Chemistry** ($n = 11$): Stoichiometry, equilibrium, and acid-base chemistry problems requiring domain-specific knowledge application.
- **Economics** ($n = 10$): Microeconomic optimization, market analysis, and financial calculations testing applied quantitative reasoning.
- **Statistics** ($n = 9$): Probability calculations, hypothesis testing, and descriptive statistics problems requiring statistical reasoning.
- **Biology** ($n = 9$): Genetics, population dynamics, and biochemical pathway problems combining qualitative and quantitative reasoning.
- **Calculus** ($n = 9$): Differentiation, integration, and optimization problems testing advanced mathematical reasoning.

TABLE 4. Distribution of problems across categories and difficulty levels.

Category	Easy	Medium	Hard	Total
Physics	3	5	3	11
Mathematics	4	5	2	11
Chemistry	3	6	2	11
Economics	4	4	2	10
Statistics	3	4	2	9
Biology	3	4	2	9
Calculus	3	4	2	9
Optimization	2	4	3	9
Total	**25**	**36**	**18**	**79**

- **Optimization** ($n = 9$): Constrained and unconstrained optimization problems requiring multi-step analytical reasoning.

2) DIFFICULTY STRATIFICATION

Problems are stratified into three difficulty levels based on cognitive complexity and solution length:

- **Easy** ($n = 25$): Single-concept problems requiring direct formula application or one-step reasoning. Typical solution length: 2–3 steps.
- **Medium** ($n = 36$): Multi-concept problems requiring integration of multiple principles or intermediate calculations. Typical solution length: 3–5 steps.
- **Hard** ($n = 18$): Complex problems requiring advanced reasoning, multiple solution stages, or sophisticated mathematical manipulation. Typical solution length: 5+ steps.

Table 4 presents the distribution of problems across categories and difficulty levels.

3) PROBLEM FORMAT

Each problem in the corpus is represented as a structured record containing:

`problem`: The natural language problem statement.
`solution`: The reference final answer.
`solution_steps`: An ordered list of reasoning steps constituting the reference solution trace.
`category`: The subject domain classification.
`difficulty`: The difficulty level (Easy, Medium, Hard).

Reference solutions were authored by domain experts and validated for correctness. Solution steps were decomposed to represent atomic reasoning operations, enabling fine-grained step-level evaluation.

C. TRANSFORMATION IMPLEMENTATION

Metamorphic transformations were implemented using a combination of rule-based methods and LLM-assisted generation, with manual validation to ensure semantic preservation.

1) TRANSFORMATION GENERATION

a: IDENTITY

The identity transformation trivially returns the original problem statement unchanged.

b: PARAPHRASE

Paraphrases were generated using GPT-4 [1] with the following prompt template:

```
Paraphrase the following problem while
preserving all numerical values, variable
names, and the exact meaning. Use
different words and sentence structures
but do not change any quantities or
add/remove information.
Original: {problem}
Paraphrased:
```

Generated paraphrases were manually reviewed to ensure semantic equivalence, with approximately 8% requiring minor corrections.

c: REORDER FACTS

For problems containing multiple independent facts, we identified fact boundaries using syntactic parsing and generated all valid permutations, selecting one uniformly at random. Problems with fewer than two independent facts received a null transformation (equivalent to identity).

d: EXPAND

Expansions were generated by augmenting problems with one or more of the following elements: (1) restatement of the question in alternative phrasing, (2) definitional clarifications for technical terms, (3) explicit statement of assumptions implicit in the original, and (4) contextual framing sentences. The prompt template emphasized that no new information relevant to the solution should be introduced.

e: CONTRACT

Contractions were generated by removing: (1) redundant qualifiers and modifiers, (2) restatements or clarifications, (3) contextual framing not essential to problem specification, and (4) verbose phrasing replaceable with concise alternatives. Manual validation ensured that all information necessary for problem solution was retained.

f: ACADEMIC CONTEXT

Academic framing was added by prepending contextual elements such as "[Exam Question]", "Consider the following problem from a physics course:", or "In a laboratory exercise, students are asked to:". The core problem statement remained unchanged.

g: BUSINESS CONTEXT

Business framing recontextualized problems within professional scenarios. For example, a physics problem about velocity might be framed as a logistics scenario, or a chemistry problem about concentration might be framed as a quality control task. Care was taken to ensure that the mathematical structure remained identical.

h: CONTRASTIVE

Contrastive augmentations added explicit contrasts with related but distinct scenarios. These took the form of clarifying statements such as "Note that unlike [related scenario], this problem involves [distinguishing feature]" or "This should not be confused with [alternative case] where [different condition applies]."

2) VALIDATION PROTOCOL

All transformations underwent a two-stage validation process:

1) **Automated checks**: Numerical values and key technical terms were extracted from original and transformed problems using regular expressions and named entity recognition, with automated flagging of discrepancies.
2) **Manual review**: A domain expert reviewed each transformation to verify semantic preservation, correct any errors introduced during generation, and ensure naturalness of the transformed text.

The validation process resulted in rejection and regeneration of approximately 12% of initial transformations, primarily due to inadvertent introduction of new information (expansion), removal of essential details (contraction), or subtle meaning shifts (paraphrase).

D. INFERENCE CONFIGURATION

1) GENERATION PARAMETERS

To account for stochastic variation in model outputs, we generated multiple solutions for each problem-transformation pair. Table 5 summarizes the inference configuration.

TABLE 5. Inference configuration parameters.

Parameter	Value
Samples per problem	3
Temperature	0.7
Top-p (nucleus sampling)	0.95
Maximum tokens	1,024
System prompt	Standard reasoning instruction

The temperature setting of 0.7 balances diversity with coherence, enabling observation of natural output variation while avoiding degenerate responses. Three samples per problem-transformation pair were collected, with metrics computed as averages over samples to reduce noise from sampling variance.

2) PROMPT FORMAT

Models were prompted using a standardized format emphasizing step-by-step reasoning:

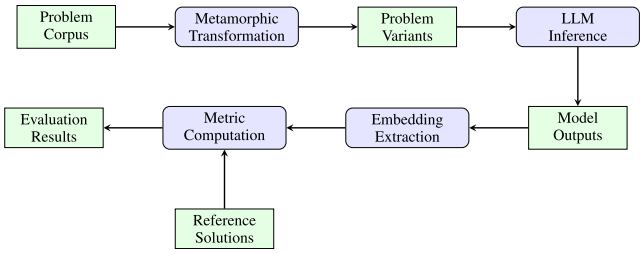


FIGURE 2. Evaluation pipeline overview. Problems undergo metamorphic transformation, model inference generates solutions, embeddings enable semantic comparison, and metrics quantify invariance relative to reference solutions.

```
Solve the following problem step by
step. Show your reasoning clearly
before providing the final answer.
Problem: {problem_text}
Solution:
```

This format encourages explicit reasoning traces amenable to step-level evaluation, while remaining neutral with respect to the specific transformations being tested.

3) COMPUTATIONAL RESOURCES

Model inference for both Hermes-4-70B and DeepSeek-R1-0528 was conducted via the Nebius AI platform, which provides cloud-based serving infrastructure for large language models.

E. EVALUATION PIPELINE

Figure 2 illustrates the end-to-end evaluation pipeline.

1) EMBEDDING COMPUTATION

Semantic embeddings were computed using the Sentence-Transformers library [48] with the `all-MiniLM-L6-v2` model. This model maps text sequences to 384-dimensional dense vectors optimized for semantic similarity tasks. Embeddings were computed for:

- Complete model solutions (for solution-level metrics)
- Individual reasoning steps (for step-level metrics)
- Concatenated reasoning traces (for trace similarity)
- Reference solutions and steps (for comparison baselines)

2) METRIC AGGREGATION

Metrics were aggregated at multiple levels to enable analysis at varying granularities:

- **Instance-level**: Raw metrics for each (problem, transformation, sample) triple.
- **Problem-level**: Averages over samples for each (problem, transformation) pair.
- **MR-level**: Averages over problems for each metamorphic relation.
- **Category-level**: Averages stratified by subject category.
- **Difficulty-level**: Averages stratified by difficulty tier.
- **Model-level**: Global averages for overall model comparison.

3) STATISTICAL TESTING

Statistical tests were conducted using SciPy [57]. For all hypothesis tests, we report exact p-values and apply a significance threshold of $\alpha = 0.05$. Multiple comparison corrections were not applied, as our analyses are primarily exploratory rather than confirmatory; we emphasize effect sizes alongside significance tests to assess practical importance.

V. RESULTS

This section presents the empirical findings from our metamorphic testing evaluation. We organize results hierarchically: overall performance comparison (Section V-A), metamorphic relation analysis (Section V-B), category-specific patterns (Section V-C), difficulty-stratified analysis (Section V-D), robustness characterization (Section V-E), and statistical significance testing (Section V-F).

A. OVERALL PERFORMANCE COMPARISON

We begin with an aggregate comparison of the two models across all problems and transformations. Figure 3 presents three complementary views of overall performance.

1) ABSOLUTE PERFORMANCE

Hermes-4-70B achieves substantially higher reasoning scores than DeepSeek-R1-0528 on both original problems (0.609 vs. 0.454) and transformed variants (0.577 vs. 0.425). This performance gap of approximately 0.15 points is consistent across transformation types, indicating that Hermes's advantage reflects genuine reasoning capability differences rather than differential robustness.

Both models exhibit performance degradation under transformation, with mean scores declining from original to transformed conditions. However, the magnitude of degradation is modest: Hermes decreases by 0.032 points (5.3% relative) while DeepSeek decreases by 0.029 points (6.4% relative). This suggests that both models maintain reasonable stability under semantic-preserving transformations at the aggregate level.

2) SCORE DELTA DISTRIBUTION

The center panel of Figure 3 reveals important distributional differences obscured by mean comparisons. Both distributions are approximately centered at zero, consistent with the expectation that semantic-preserving transformations should not systematically improve or degrade performance. However, DeepSeek-R1-0528 exhibits heavier tails, with more instances of extreme degradation (score delta < -0.5) and extreme improvement (score delta > 0.4).

This higher variance suggests that DeepSeek's reasoning process is more sensitive to surface-level input variations, even when aggregate performance appears stable. The practical implication is reduced predictability: users of

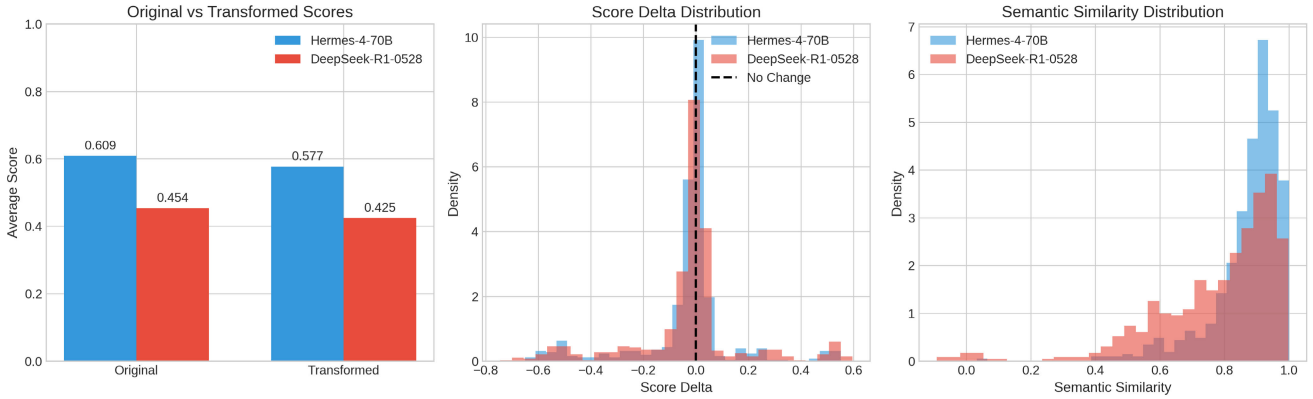


FIGURE 3. Overall performance comparison between Hermes-4-70B and DeepSeek-R1-0528. Left: Average scores on original and transformed problems, showing Hermes maintains higher absolute performance. Center: Distribution of score deltas, with both models centered near zero but exhibiting different variance patterns. Right: Semantic similarity distributions, with Hermes showing stronger concentration at high similarity values.

TABLE 6. Overall performance summary. Higher values are better for all metrics except mean absolute delta and degradation rate.

Metric	Hermes-4-70B	DeepSeek-R1-0528
Overall Average Score	0.577	0.425
Step Accuracy	0.531	0.640
Mean Score Delta	−0.032	−0.029
Mean Absolute Delta	0.0948	0.1208
Invariance Score	0.905	0.879
Degradation Rate ($\theta = 0.1$)	13.6%	17.6%
Mean Semantic Similarity	0.874	0.779
Consistency	0.883	0.862

DeepSeek-R1-0528 may encounter greater variability in response quality across semantically equivalent query formulations.

3) SEMANTIC SIMILARITY OF REASONING TRACES

The right panel of Figure 3 displays the distribution of semantic similarity between reasoning traces produced for original and transformed problems. Hermes-4-70B exhibits a strong concentration of similarity values near 0.95, indicating highly consistent reasoning processes across transformations. DeepSeek-R1-0528 shows a broader distribution with a secondary mode around 0.7 and a tail extending below 0.5.

This divergence in trace consistency has important implications for interpretability and debugging. When Hermes produces different answers for transformed problems, the reasoning traces remain largely parallel, facilitating identification of the divergence point. DeepSeek's more variable traces suggest that transformations can induce qualitatively different reasoning paths, complicating error analysis.

Table 6 summarizes key aggregate metrics.

Degradation Rate ($\theta = 0.1$) is 13.6% for Hermes and 17.6% for DeepSeek. The consistency metric is 0.883 for Hermes and 0.862 for DeepSeek, reflecting the average stability of transformed scores across metamorphic relations.

4) THE ACCURACY-FIDELITY PARADOX

A striking finding emerges from comparison of overall scores and step accuracy: while Hermes-4-70B achieves higher solution-level scores (0.577 vs. 0.425), DeepSeek-R1-0528 demonstrates superior step-level accuracy (0.640 vs. 0.531). This apparent paradox suggests fundamentally different reasoning strategies.

DeepSeek appears to execute individual reasoning steps with higher fidelity to reference solutions, yet these steps less reliably compose into correct final answers. This pattern is consistent with a reasoning process that is locally accurate but globally fragile—individual steps are correct in isolation but fail to maintain coherence across the full solution path. Hermes, conversely, may employ reasoning steps that diverge from reference formulations while nonetheless arriving at correct conclusions through alternative valid paths.

B. METAMORPHIC RELATION ANALYSIS

We next examine how different metamorphic relations affect model performance, revealing transformation-specific vulnerability patterns. Figure 4 presents a comprehensive view of score delta distributions across MRs.

1) IDENTITY BASELINE

The identity transformation establishes a baseline for intrinsic model variance. Both models show near-zero mean delta under identity (Hermes: +0.001, DeepSeek: −0.011), confirming that our evaluation methodology does not introduce systematic bias. The small negative delta for DeepSeek reflects sampling variance across the three generations per problem.

2) STRUCTURAL TRANSFORMATIONS

a: PARAPHRASE

Paraphrasing induces moderate degradation for both models, with Hermes showing a mean delta of −0.057 and DeepSeek −0.082. However, the distributions differ markedly: Hermes exhibits relatively uniform degradation

across problems, while DeepSeek shows a bimodal pattern with many problems unaffected and others severely degraded. This suggests that DeepSeek's sensitivity to paraphrase is problem-dependent, potentially related to specific lexical or syntactic patterns in the training data.

b: REORDER FACTS

Fact reordering proves particularly challenging for DeepSeek-R1-0528, which exhibits a mean delta of -0.078 compared to Hermes's -0.083. More concerning is DeepSeek's high variance under this transformation, with multiple instances of catastrophic degradation (delta < -0.5). This vulnerability suggests that DeepSeek's reasoning process may encode implicit dependencies on presentation order, violating the expected invariance to fact permutation.

3) VERBOSITY TRANSFORMATIONS

a: EXPAND

Expansion has minimal effect on both models: Hermes exhibits near-zero delta ($+0.002$) as does DeepSeek ($+0.006$).

b: CONTRACT

Contraction effects differ substantially: Hermes shows slight improvement ($+0.012$) while DeepSeek exhibits notable degradation (-0.096).

4) CONTEXTUAL TRANSFORMATIONS

a: ACADEMIC CONTEXT

Academic framing has minimal effect on either model (Hermes: $+0.014$, DeepSeek: -0.009), indicating robust reasoning regardless of scholarly contextual framing.

b: BUSINESS CONTEXT

Business framing similarly shows modest effects (Hermes: $+0.002$, DeepSeek: $+0.044$), with both models exhibiting slight improvement. This may reflect exposure to business-contextualized problems in training data, or could indicate that concrete applied framing facilitates reasoning compared to abstract formulations.

c: CONTRASTIVE

Contrastive transformations induce the largest degradation for Hermes-4-70B (mean delta -0.147), representing a substantial and consistent vulnerability. DeepSeek shows smaller mean degradation (-0.005) but high variance. The contrastive transformation's disruptive effect suggests that both models struggle to maintain focus on the target problem when presented with explicit contrasts to alternative scenarios. This has practical implications for prompt engineering: including comparative information intended to clarify may paradoxically degrade reasoning quality.

5) CROSS-MODEL MR COMPARISON

Figure 5 provides a detailed view of score delta patterns across the MR $\times$ Category space for both models, enabling identification of specific vulnerability pockets.

Several patterns emerge from the heatmap comparison:

- **Hermes vulnerabilities**: Contrastive transformations cause consistent degradation across categories, with Statistics (-0.199), Calculus (-0.187), and Economics (-0.110) most affected. The contrastive $\times$ Mathematics cell shows particularly severe degradation (-0.221).
- **DeepSeek vulnerabilities**: Paraphrase and reorder_facts cause severe degradation in formal domains. The paraphrase $\times$ Mathematics cell shows -0.203 delta, and reorder_facts $\times$ Mathematics shows -0.202. Contract transformations also severely affect Mathematics (-0.203) and Calculus (-0.176).
- **Relative strengths**: DeepSeek shows resilience to contrastive transformations in Chemistry ($+0.182$) and Economics ($+0.074$), while Hermes maintains stability under expand across all categories.

C. CATEGORY-SPECIFIC PATTERNS

Figure 6 presents category-stratified results, revealing substantial variation in both absolute performance and transformation sensitivity across subject domains.

1) ABSOLUTE PERFORMANCE BY CATEGORY

Hermes-4-70B outperforms DeepSeek-R1-0528 across all eight categories, with the performance gap ranging from 0.06 (Biology) to 0.26 (Chemistry). The radar chart in Figure 6 (bottom-right) visualizes this consistent advantage, with Hermes's polygon strictly containing DeepSeek's across all axes.

Category-level performance rankings differ between models.

For Hermes, the ordering from highest to lowest is: Calculus (0.690) > Economics (0.658) > Mathematics (0.625) > Biology (0.618) > Physics (0.593) > Chemistry (0.587) > Statistics (0.546) > Optimization (0.462).

For DeepSeek: Calculus (0.590) > Biology (0.549) > Mathematics (0.471) > Economics (0.450) > Optimization (0.443) > Physics (0.427) > Statistics (0.426) > Chemistry (0.331).

The substantial Chemistry gap (0.587 vs. 0.331) warrants attention. Chemistry problems often require domain-specific knowledge (e.g., molecular weights, reaction stoichiometry) that may be differentially represented in training corpora. DeepSeek's poor Chemistry performance suggests potential gaps in domain coverage.

2) TRANSFORMATION SENSITIVITY BY CATEGORY

The top-right panel of Figure 6 reveals category-specific transformation sensitivity patterns. Key observations:

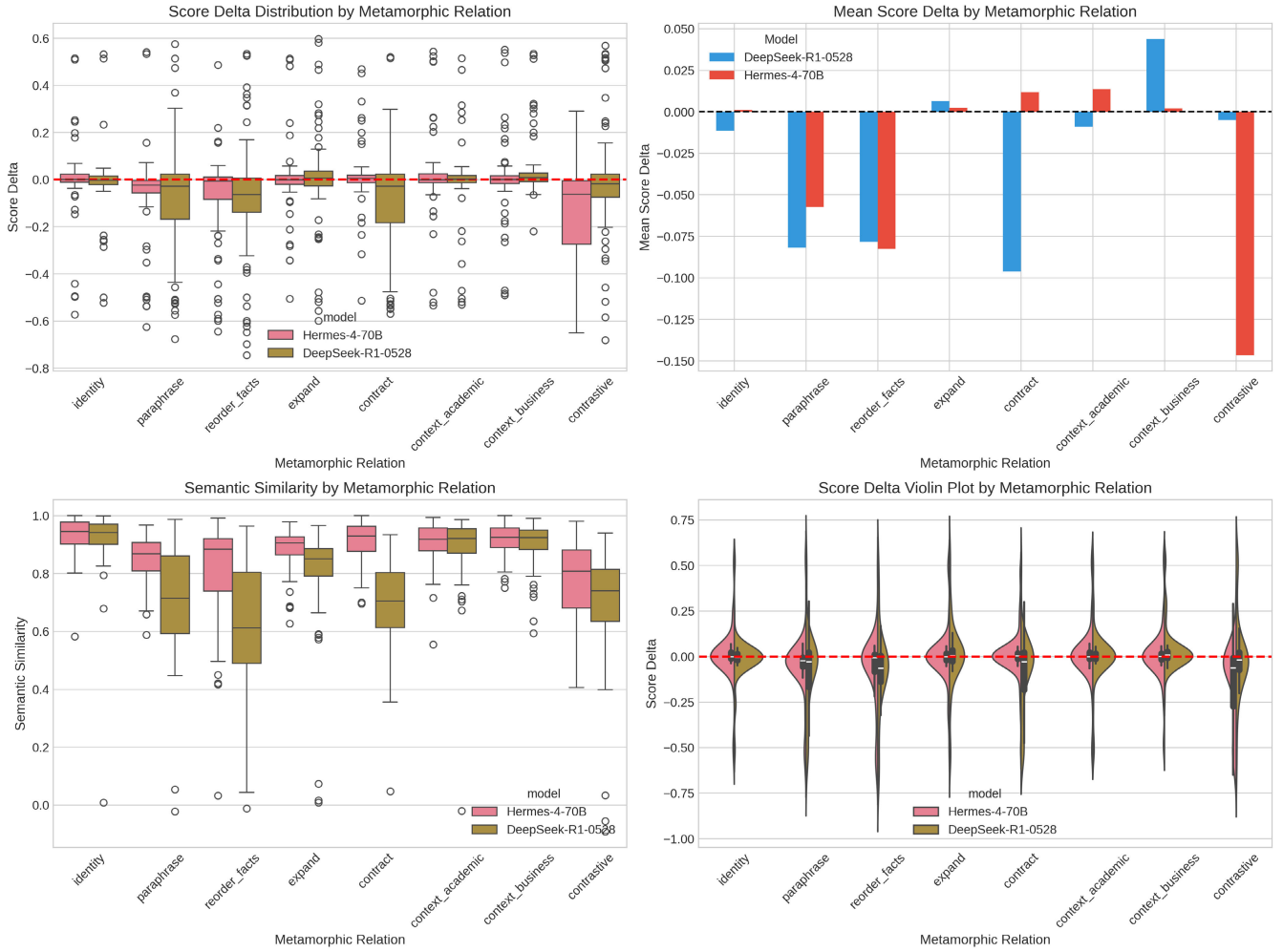


FIGURE 4. **Metamorphic relation analysis. Top-left: Box plots of score delta by MR, showing systematic differences across transformations. Top-right: Mean score delta comparison, with contrastive transformations inducing largest degradation. Bottom-left: Semantic similarity distributions by MR, revealing paraphrase and reorder_facts as most disruptive for DeepSeek. Bottom-right: Violin plots showing full distributional shapes.**

- **Biology**: Both models show degradation, with DeepSeek slightly worse (mean delta ≈ -0.06 vs. ≈ -0.04 for Hermes).
- **Chemistry**: DeepSeek shows positive mean delta ($\approx +0.04$), suggesting transformations may actually help by rephrasing problems into more recognizable patterns. Hermes is essentially neutral (mean delta $\approx +0.01$).
- **Calculus**: Both models exhibit degradation of similar magnitude (Hermes ≈ -0.10, DeepSeek ≈ -0.10). In combination with the MR-level analysis, this highlights that DeepSeek's largest vulnerabilities in Calculus arise from specific transformations such as paraphrase and fact reordering rather than a uniformly lower baseline.
- **Statistics**: Hermes shows notable degradation (mean delta ≈ -0.09), driven primarily by contrastive transformations, while DeepSeek degrades more moderately (mean delta ≈ -0.05).

The box plot distributions (bottom-left) reveal that category-level means can obscure substantial within-category variance. Calculus and Statistics show particularly wide interquartile ranges for both models, indicating problem-specific factors beyond category membership influence transformation sensitivity.

D. DIFFICULTY-STRATIFIED ANALYSIS

Figure 7 presents results stratified by problem difficulty, testing the hypothesis that complex problems exhibit greater transformation sensitivity.

1) PERFORMANCE DEGRADATION WITH DIFFICULTY

Both models show expected performance degradation as problem difficulty increases.

Hermes-4-70B scores decrease from 0.689 (Easy) to 0.586 (Medium) to 0.499 (Hard), representing a 27.6% relative decline. DeepSeek-R1-0528 shows: 0.559 (Easy) to 0.410 (Medium) to 0.408 (Hard).

The convergence of DeepSeek's Medium and Hard performance suggests a performance floor—once problems exceed a certain complexity threshold, additional difficulty does not

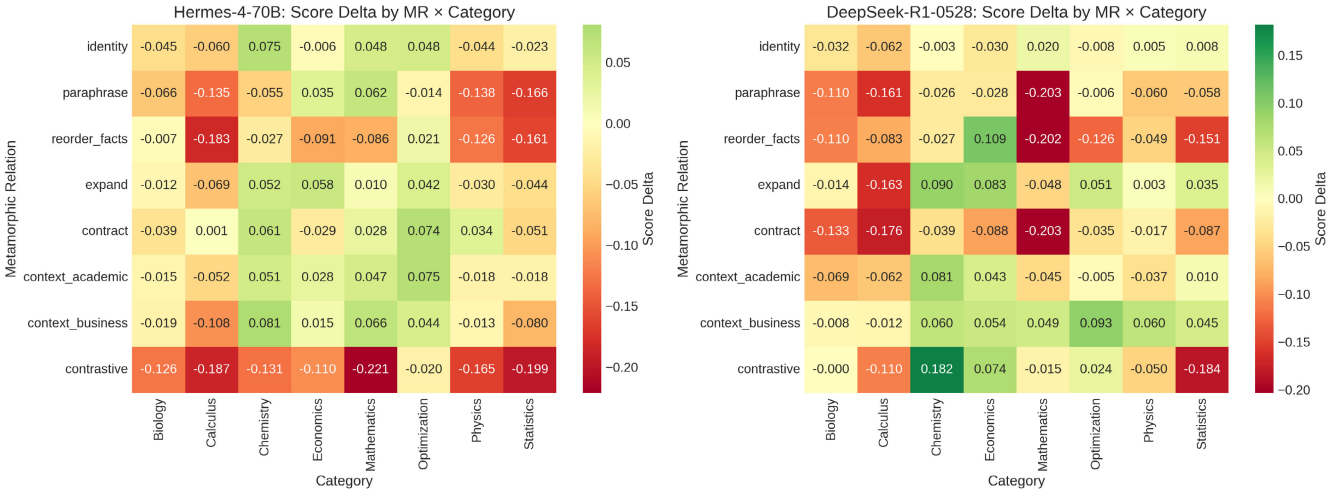


FIGURE 5. Score delta heatmaps by metamorphic relation and category. Left: Hermes-4-70B shows relatively uniform patterns with localized vulnerabilities (contrastive across multiple categories). Right: DeepSeek-R1-0528 exhibits more severe localized degradation, particularly for paraphrase and reorder_facts in Calculus and Mathematics.

TABLE 7. Statistical significance tests comparing Hermes-4-70B and DeepSeek-R1-0528. All tests are two-tailed with $\alpha = 0.05$.

Metric	Test	Statistic	p-value	Effect Size
Score Delta	Mann-Whitney U	151,330	0.767	$d = -0.03$ (negligible)
Semantic Similarity	Mann-Whitney U	262,618	< 0.0001	$d = 0.60$ (medium)
MR Effect (Combined)	Kruskal-Wallis H	87.25	< 0.0001	—

further degrade DeepSeek's performance. This could indicate either consistent failure modes for complex problems or, more optimistically, robust (if limited) reasoning capabilities for challenging tasks.

2) TRANSFORMATION SENSITIVITY BY DIFFICULTY

The bottom-left panel of Figure 7 reveals an intriguing divergence in how difficulty modulates transformation sensitivity:

- **Hermes-4-70B** shows monotonically increasing degradation with difficulty: Easy (-0.020) $\rightarrow$ Medium (-0.035) $\rightarrow$ Hard (-0.041). This pattern aligns with intuition: more complex reasoning chains provide more opportunities for transformation-induced perturbations to compound.
- **DeepSeek-R1-0528** shows a non-monotonic pattern: Easy (-0.056) $\rightarrow$ Medium (-0.012) $\rightarrow$ Hard (-0.025). The high sensitivity at Easy difficulty is surprising and may reflect overconfidence on simple problems that is disrupted by transformations.

The interaction between difficulty and MR type is further elaborated in Figure 8, which presents the full MR $\times$ Difficulty breakdown.

Key observations from the comprehensive heatmaps:

- **DeepSeek Easy-level vulnerabilities**: Contract (-0.193) and paraphrase (-0.142) cause severe degradation at Easy difficulty, while Medium and Hard problems show more moderate effects. This suggests that simple problems may be solved via pattern matching that is disrupted by surface-level changes.
- **Hermes contrastive consistency**: Contrastive degradation is consistent across difficulty levels (-0.168, -0.134, -0.142 for Easy, Medium, Hard), indicating a systematic vulnerability independent of problem complexity.
- **Semantic similarity patterns**: Hermes maintains high trace similarity (> 0.85) across most MR-difficulty combinations. DeepSeek shows reduced similarity for paraphrase and reorder_facts, particularly at Medium difficulty, indicating reasoning path disruption.

E. ROBUSTNESS CHARACTERIZATION

We synthesize the preceding analyses into an overall robustness characterization, examining problem-level stability and cumulative performance distributions.

1) PROBLEM-LEVEL STABILITY

Figure 9 examines robustness at the individual problem level, quantifying consistency of model performance across transformation variants.

For each problem, we compute the delta range—the difference between maximum and minimum score delta across all MR variants. A stable model should exhibit low delta range, indicating consistent performance regardless of transformation type.

The left panel of Figure 9 shows that Hermes-4-70B exhibits a mode near 0.1 with rapid decay, while DeepSeek-R1-0528 shows a broader distribution extending beyond 0.6. This confirms that Hermes provides more predictable performance across problem formulations.

Category-level analysis (right panel) reveals that:

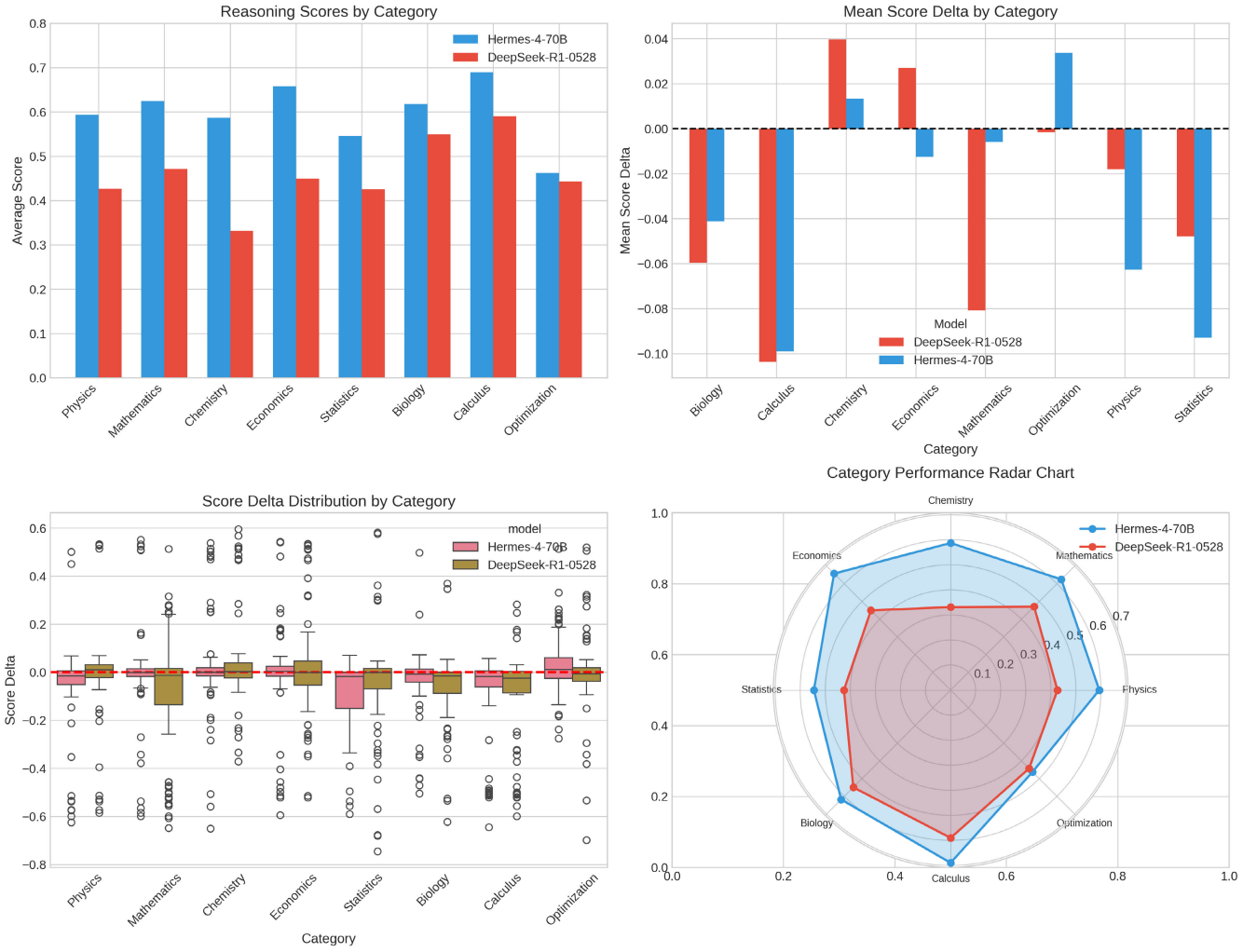


FIGURE 6. Category-wise analysis. Top-left: Reasoning scores by category showing Hermes advantage across all domains, with largest gaps in Chemistry and Economics. Top-right: Mean score delta by category revealing domain-specific transformation sensitivity. Bottom-left: Box plots of score delta distribution by category. Bottom-right: Radar chart providing holistic comparison of category-level performance.

- Chemistry shows the largest stability gap, with DeepSeek's mean delta range (0.48) approximately 27% higher than that of Hermes (0.38).
- Physics and Calculus are the only categories where DeepSeek shows greater stability than Hermes, though both models exhibit relatively low delta ranges in these domains.
- Statistics and Optimization show high instability for both models, suggesting inherent difficulty in maintaining consistent reasoning for these problem types.

2) CUMULATIVE PERFORMANCE ANALYSIS

The cumulative distribution of score delta (see Figure 10) provides a comprehensive robustness comparison. Key observations:

- At the -0.2 threshold, approximately 11% of Hermes instances fall below versus 14% for DeepSeek, indicating lower tail risk for Hermes.
- The curves cross near $\Delta = 0$, with both models showing approximately 55–57% of instances below zero delta.
- In the positive delta region, DeepSeek shows slightly more instances of substantial improvement ($\Delta > 0.2$), consistent with its higher variance profile.

F. STATISTICAL SIGNIFICANCE

Table 7 summarizes statistical tests comparing the two models across key metrics.

The Mann-Whitney U test reveals that score delta distributions do not differ significantly between models ($p = 0.767$), with a negligible effect size ($d = -0.03$). This indicates that, on average, both models exhibit comparable levels of score degradation under metamorphic transformations.

In contrast, semantic similarity shows a substantial and statistically significant difference ($p < 0.0001$, $d = 0.60$), confirming that Hermes maintains more consistent

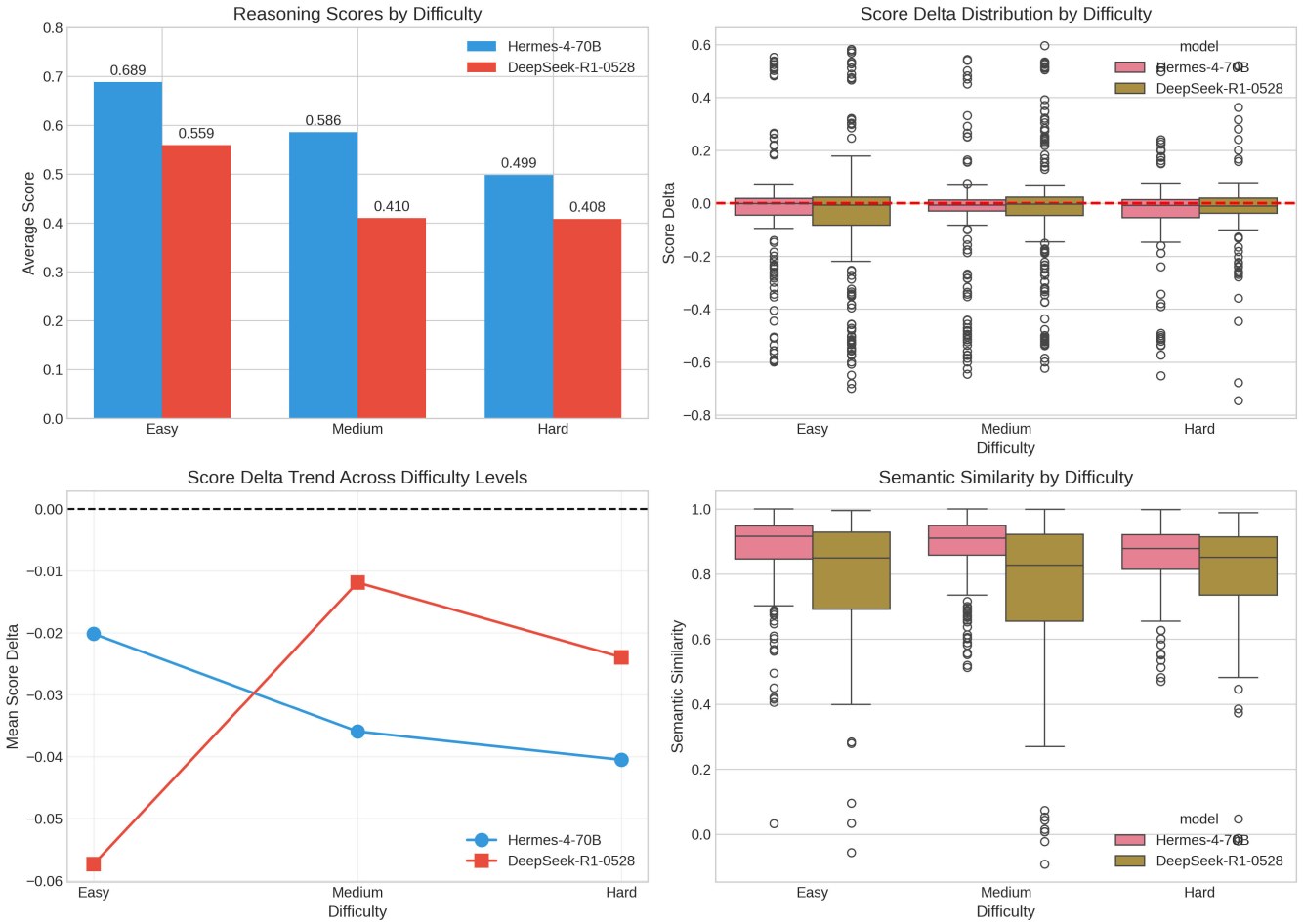


FIGURE 7. Difficulty-wise analysis. Top-left: Reasoning scores decrease with difficulty for both models, with Hermes maintaining larger advantage at harder levels. Top-right: Score delta distributions by difficulty showing similar patterns across levels. Bottom-left: Mean score delta trends, with Hermes showing monotonic degradation increase while DeepSeek shows non-monotonic pattern. Bottom-right: Semantic similarity remains relatively stable across difficulty levels.

reasoning traces across transformations despite similar score-level performance.

The Kruskal-Wallis test confirms that score deltas differ significantly across MR types ($p < 0.0001$), validating that different transformations induce meaningfully different effects rather than uniform random perturbations.

1) MR-SPECIFIC COMPARISONS

Table 8 presents pairwise model comparisons for each metamorphic relation.

Three MRs show statistically significant between-model differences:

- **Contract**: DeepSeek shows significantly greater degradation ($p = 0.001$), with a medium effect size ($d = 0.59$). Hermes maintains near-baseline performance ($\mu = +0.012$) while DeepSeek exhibits substantial degradation ($\mu = -0.096$).
- **Business**: DeepSeek shows a small but significant positive effect ($p = 0.047$, $d = -0.29$), suggesting the business context framing slightly benefits DeepSeek more than Hermes.

TABLE 8. Statistical comparison by metamorphic relation. Significant differences ($p < 0.05$) are indicated in bold.

MR	Hermes μ	DeepSeek μ	U-stat	p-value	Cohen's d
Identity	+0.001	−0.012	3,434	0.276	0.09
Paraphrase	−0.057	−0.082	3,109	0.970	0.12
Reorder Facts	−0.083	−0.078	3,556	0.130	−0.02
Expand	+0.002	+0.006	2,815	0.289	−0.02
Contract	+0.012	**−0.096**	4,076	**0.001**	0.59
Academic	+0.014	−0.009	3,167	0.873	0.14
Business	+0.002	**+0.044**	2,548	**0.047**	−0.29
Contrastive	**−0.147**	−0.005	2,251	**0.003**	−0.62

- **Contrastive**: Hermes shows significantly greater degradation ($p = 0.003$), with a medium-to-large effect size ($d = -0.62$). Hermes exhibits substantial degradation ($\mu = -0.147$) while DeepSeek remains near baseline ($\mu = -0.005$).

These findings identify contraction sensitivity as a distinctive DeepSeek weakness and contrastive sensitivity as a distinctive Hermes weakness, providing actionable guidance for model selection based on expected input characteristics.

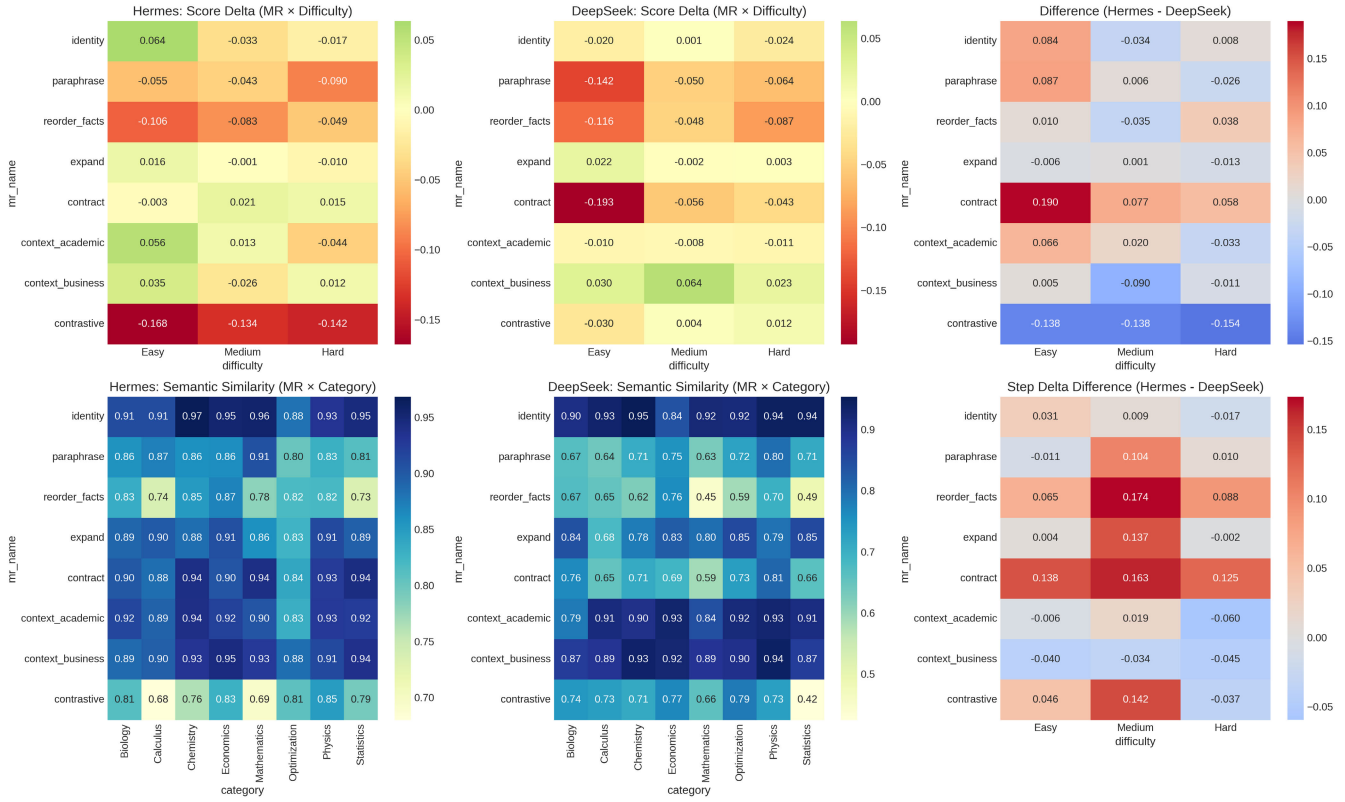


FIGURE 8. Comprehensive heatmaps showing MR × Difficulty interactions. Top row: Score delta for Hermes (left), DeepSeek (center), and their difference (right). Bottom row: Semantic similarity for Hermes (left) and DeepSeek (center), with step delta difference (right). Notable patterns include DeepSeek's severe contract degradation at easy difficulty and Hermes's contrastive vulnerability across all difficulty levels.

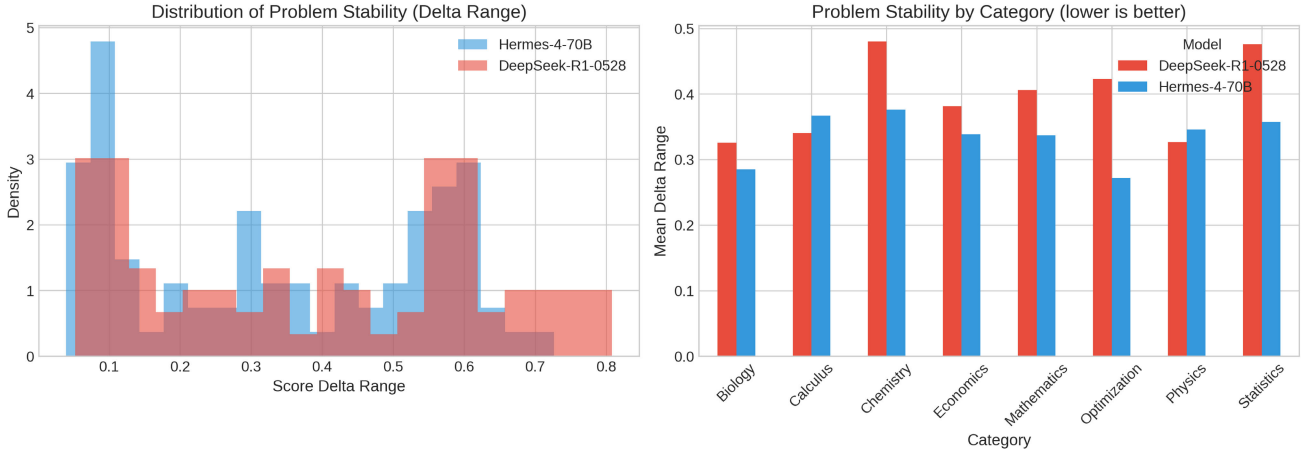


FIGURE 9. Problem-level stability analysis. Left: Distribution of delta range (max delta − min delta across MRs) for each problem, with Hermes showing concentration at lower values indicating greater stability. Right: Mean delta range by category, with Hermes more stable across all categories except Physics.

2) CORRELATION ANALYSIS

We examine the relationship between score delta and semantic trace similarity:

- **Hermes-4-70B**: Pearson $r = 0.31$ ($p < 0.0001$), Spearman $\rho = 0.26$ ($p < 0.0001$).
- **DeepSeek-R1-0528**: Pearson $r = 0.40$ ($p < 0.0001$), Spearman $\rho = 0.42$ ($p < 0.0001$).

The positive correlations indicate that score degradation is associated with reasoning trace divergence for both models. Interestingly, the relationship is stronger for DeepSeek, suggesting that DeepSeek's performance changes are more predictable from trace analysis. When DeepSeek's reasoning traces diverge from the original, score degradation tends to follow more reliably. In contrast, Hermes can maintain more stable traces even when

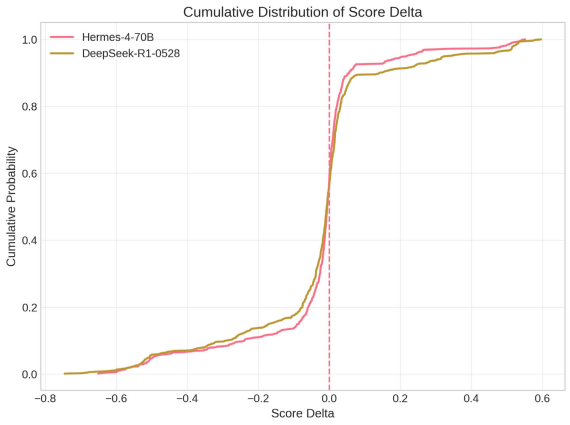


FIGURE 10. Cumulative distribution of score delta for both models. Hermes-4-70B (pink) shows a curve shifted slightly right relative to DeepSeek-R1-0528 (olive), indicating lower tail risk.

scores change, indicating a degree of decoupling between surface-level reasoning expression and underlying solution quality.

VI. CONCLUSION

This paper introduced a metamorphic testing framework for evaluating semantic invariance in large language model reasoning. By systematically applying eight metamorphic relations—spanning structural, verbosity, and contextual transformation categories—we assessed the robustness of LLM reasoning processes to semantically equivalent input variations. Our comparative analysis of Hermes-4-70B and DeepSeek-R1-0528 across 79 reasoning problems yielded several findings that advance understanding of LLM reasoning capabilities and limitations.

A. PRINCIPAL FINDINGS

Our evaluation revealed a striking dissociation between solution accuracy and reasoning fidelity: Hermes-4-70B achieves superior overall scores (0.577 vs. 0.425) while DeepSeek-R1-0528 demonstrates higher step-level accuracy (0.640 vs. 0.531). This paradox suggests fundamentally different reasoning strategies that aggregate metrics fail to distinguish. Both models achieve similar invariance scores ($\sim$ 0.90), yet manifest distinct vulnerability profiles—DeepSeek exhibits catastrophic sensitivity to paraphrase and fact reordering in formal domains, while Hermes shows consistent degradation under contrastive transformations across all categories. Semantic similarity analysis confirms that Hermes maintains more coherent reasoning traces under transformation (0.874 vs. 0.779), indicating greater stability in underlying reasoning processes.

B. METHODOLOGICAL CONTRIBUTIONS

We contribute a comprehensive framework for semantic invariance evaluation that complements existing accuracy-based benchmarks. Our metamorphic relation taxonomy provides a standardized vocabulary for characterizing transformation types, enabling systematic comparison across models and studies. The multi-level evaluation approach—assessing solution quality, step accuracy, and trace coherence—offers a more complete picture of reasoning robustness than any single metric provides.

C. LIMITATIONS

Several limitations of this study warrant acknowledgment. First, the two evaluated models exhibit substantially different baseline performance levels (Hermes: 0.609, DeepSeek: 0.454 on original problems), which may complicate between-model comparisons. While our delta-based analysis is designed to isolate transformation effects from absolute performance, ceiling and floor effects could influence the magnitude of observable degradation, a high-performing model has more room to degrade, while a lower-performing model may already operate near its performance floor for certain problem types. Additionally, the statistical tests employed (Mann-Whitney U, Kruskal-Wallis) assume sample independence, whereas the experimental design involves paired observations, the same problems are evaluated across both models and across metamorphic relations. While paired tests (e.g., Wilcoxon signed-rank, Friedman) might be more formally appropriate, we note that our analysis emphasizes effect sizes over p-values, and the delta-based evaluation already captures within-problem variation. The reported effect sizes and observed patterns remain informative for practical interpretation.

D. FUTURE OUTLOOK

This work opens several avenues for future research. Extending the evaluation to a broader model set would establish the generality of observed patterns. Expanding the transformation taxonomy would enable finer-grained robustness characterization. Most ambitiously, our diagnostic framework could guide the development of training methodologies that explicitly optimize for semantic invariance, advancing toward language models whose reasoning is genuinely robust to the inevitable variability of real-world inputs.

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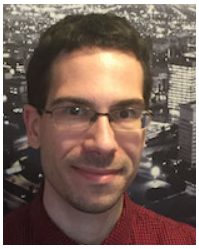
During the preparation of this work, the author(s) used ChatGPT/Copilot/Gemini/Claude for code completion and pseudocode generation, Grammarly for English language editing, and Mendeley for bibliography management. They reviewed and edited the content as needed and take(s) full responsibility for the content of this publication. They would like to thank the BARCELONA Supercomputing Center for providing access to MareNostrum 5 and technical support throughout this research.

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